

ROBBY ULUNG PAMBUDI

Software Engineering

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SUMMARY

Software Engineer with 4+ years of experience building scalable backend, AI-driven, and cloud-native applications. Experienced in developing Retrieval-Augmented Generation (RAG) systems, LLM-powered applications, and distributed architectures using FastAPI, LangChain, ChromaDB, Docker, and modern cloud technologies. Proven track record of leading cross-functional teams, optimizing system performance by up to 60%, and delivering production systems serving over 250,000 users in SPMB Jatim 2024–2025. Passionate about applying Large Language Models and Natural Language Processing to build intelligent systems that solve real-world problems.

EDUCATION

Sepuluh Nopember Institute of Technology Feb 2026 – Present
Master of Informatics Surabaya, East Java

- Professor’s Research Assistant – Distributed Systems.
- Teaching Assistant for Microservices and Application Development.

Sepuluh Nopember Institute of Technology Sep 2021 – Aug 2025
Bachelor of Informatics - GPA 3.61 Surabaya, East Java

- Data Structure’s, Fundamental Programming, and Operation System Teaching Assistant
- Finalists of SSG x Hackfest 2024 held by Ciputra University

EXPERIENCE

Techflouu  Dec 2025 – Present
Fullstack Engineer - Full Time Singapore, Remote

- Designed and implemented a scalable multi-tenant architecture supporting 10+ tenants with isolated data and configuration.
- Optimized multi-tenant queries and data access patterns, reducing response time by up to 40% and improving system reliability.

Techflouu  Aug – Dec 2025
Fullstack Developer - Internship Singapore, Remote

- Presented technical solutions and project progress to clients, ensuring clear communication and alignment with business requirements.
- Refactored over 20 legacy modules, improving code maintainability and reducing technical debt across the application.

SPMB Jawa Timur  Jan 2024 – Jul 2025
Fullstack Developer - Full Time Surabaya, Remote

- Optimized frontend query performance, improving loading and interaction times by 60%, enhancing user experience across both student and administrative portals
- Led the development and continuous improvement of core features such as enrollment, re-enrollment, and score reporting for a system used by over 250k students across East Java

Filin Movie  Jul 2023 – Feb 2024
Frontend Developer - Intern Surabaya, Remote

- Designed a solution to protect movies from illegal downloading and screen sharing using DRM (Digital Rights Management) that resulted in an 80% reduction in piracy tested using 8 different browsers.

Aseven Indonesia  Dec 2022 – Jul 2023
Frontend Developer - Freelance Surabaya, Remote

- Developed core features for *Spectra*, a web-based platform supporting internal operations of PLN, Indonesia’s state-owned electricity company.
- Implemented 13 out of 30+ key pages, including ULP, UP3, Employee, and Parent dashboards, ensuring responsive and maintainable UI components.

EMR Berkah Sehat  Jan – Apr 2023
Frontend Developer - Freelance Surabaya, Remote

- Built a scalable HOC for RBAC in a medical app, enabling dynamic route protection using a role-permission matrix with 5 roles and 20+ rules.

SELECTED PROJECTS

RAGForge: Retrieval-Augmented Generation template  | [Python](#), [LangChain](#), [ChromaDB](#) Feb 2025 – Jul 2025

- A production-ready template for building Retrieval-Augmented Generation (RAG) systems using **FastAPI**, **ChromaDB**, and **LangChain**.
- Built for scalability and maintainability using **dependency injection** and Docker-based deployment.

DiLLeMA: Distributed LLM Computing Framework  | [Python](#), [Ray](#), [vLLM](#) Feb 2025 – Jul 2025

- A final-year thesis project to build an extensible framework for distributed LLM inference over GPU clusters using **Ray** for workload distribution and **vLLM** as the core inference engine.
- Supports modular integration with RAG pipelines and focuses on high-throughput, scalable deployments for LLM workloads.

Visualization of Sorting Algorithm  | [Typescript](#), [D3.JS](#), [NextJs](#) Feb - Mar 2022

- An algorithm visualization website that provides an overview of the movement of 17 sorting algorithms was created to make it easier for everyone to understand the algorithm.

ACTIVITIES

Team Research Flood Monitoring and Early Warning System (DigiFlood) <i>Lead Engineer</i>	Feb 2026 – Present <i>Offline</i>
- Led the end-to-end development of DigiFlood, a flood monitoring and early warning system, coordinating cross-functional teams across backend, frontend, and mobile platforms. - Integrated YOLO-based object detection systems into drone platforms and collaborated across multiple divisions to align AI models, hardware integration, and operational requirements for successful deployment.	
Team Research Indonesian Sustainable Palm Oil (ISPO) <i>Software Engineer</i>	Maret 2025 – Jul 2025 <i>Offline</i>
- Conducted in-depth research on sustainable palm oil certification, supporting the alignment of ISPO standards with 10+ regulatory references. - Designed and developed a digital certification system tailored for 5k+ users across farmer groups, cooperatives (KUD), Internal Control Systems (ICS), survey institutions, and provincial/regency government offices	
WasteTrack 2024 <i>Software Engineer</i>	Aug 2024 – Maret 2025 <i>Offline</i>
Led the design and development of a web-based system using NextJS for the frontend, and Golang for the backend, supporting over 100 participants from 34 provinces in the national WasteTrack 2024 competition	
Bangkit Academy <i>Cloud Computing Learning Path</i>	Feb - Jul 2024 <i>Online</i>
Learned and completed a comprehensive cloud computing course, completing a total of 98% of the course on time	
ITS Software Engineering Laboratory <i>Lab Assistant</i>	Jul - Des 2023 <i>Offline</i>
Responsible for being a liaison between customers and the laboratory in terms of creating a website application and has successfully obtained contracts with 5 clients.	
ITS Expo 2023 <i>Head of IT Division</i>	Dec 2022 – Sep 2023 <i>Offline</i>
Managed and led a team of 27 staff to effectively collaborate in solving tasks, achieving a 92% completion rate for the division's KPIs.	

PUBLICATION

- Pambudi, R., et al.**, "DiLLeMa: An extensible and scalable framework for distributed large language models (LLMs) inference on multi-GPU clusters," *SoftwareX*, Elsevier, 2026, doi:10.1016/j.softx.2026.102537.
- Pambudi, R., et al.**, "A Novel Approach to Image Steganography Using Pixel Differencing for Better Data Hiding," *Proceedings of the IEEE Conference on ONCON*, Beijing, China, Dec. 8–10, 2024, pp. 123–130, doi:10.1109/ONCON62778.2024.10931677.

TECHNICAL SKILLS

Languages: Python, PHP, C, C++, HTML, CSS, Bash, JavaScript, Typescript, SQL, NoSQL
Developer Tools: VS Code
Technologies/Frameworks: Linux, Git, GitHub, NGINX, NextJS, NodeJS, ExpressJS, MySQL, PostgreSQL, MongoDB, Docker, D3.js, ReactJS